MIL-T-82381(TD)
4 June 1969
SUPERSEDING
NAVTRADEVCEN 3313-534
22 April 1968

MILITARY SPECIFICATION

TRAINING AID; MOCKUP, GYRO DEMONSTRATOR, PENDULOUS INTEGRATING GYRO ACCELEROMETER (PIGA)

This specification has been approved by the Naval Training Device Center, Department of the Navy.

1. SCOPE

1.1 This specification establishes the requirements for the production of a three-dimensional mockup gyro demonstrator which will serve as a classroom training aid to demonstrate visually, the principles of operation, components arrangement, and the identification of parts in a Pendulous Integrating Gyro Accelerometer (PIGA).

1.1.1 This item has been designated by Naval Training Device Center, as Device 6D7 for Navy procurement.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on the date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

Federal

PPP-B-601	Boxes, Wood, Cleated-plywood
PPP-B-621	Boxes, Wood, Nailed and Lock-corner
PPP-B-636	Box, Fiberboard

Military

MIL-P-116	Preservation; Methods of
MIL-A-8625	Anodic Coatings, For Aluminum and Aluminum Alloys
MIL-L-10547	Liners, Case and Sheet, Overwrap; Water-Vaporproof or Waterproof, Flexible

FSC 6910

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MIL-T-23991	Training Devices, Military; General Specification for

STANDARDS

Military

MIL-STD-129	Marking for Shipment and Storage

DRAWINGS

Naval Training Device Center (NAVTRADEVCEN)

3247-040114	Model Configuration PIGA

PHOTOGRAPHS

Naval Training Device Center (NAVTRADEVCEN)

BXA-6983-8-61	Device 6D7, Pendulous Integrating Gyro Accelerometer

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the Procuring Contracting Officer.)

3. REQUIREMENTS

3.1 Materials.- Materials shall be as specified in 3.1.1 through 3.1.5.1. Materials not specified herein shall conform to the requirements of MIL-T-23991.

3.1.1 Materials used in the construction of the training aid shall be of the highest commercial quality obtainable, of the lightest practical weight, and suitable for the use intended.

3.1.2 Metals.- Metal parts shall be of a corrosion-resistant metal or of a metal given a corrosion-resistant treatment or coating in accordance with MIL-T-23991.

3.1.3 Wood.- Wood used shall be of select grades of hardwood lumber, having good appearance and finishing qualities, suitable for either clear or opaque finishes. Wood shall be properly seasoned, free from checks, splits and crossgrain or seasoning defects which may cause warping.

3.1.4 Glue.- Glue used shall not be adversely affected by lacquer, paint, or by the environmental conditions specified in 3.3.

3.1.5 Plastics.- Plastics shall be durable and resistant to cracking, brittleness, discoloration, deformation and similar defects, upon aging or when subjected to environmental conditions specified in 3.3.

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3.1.5.1 Plastic cementing.- Contact surfaces of plastic pieces to be joined shall be machined or sanded. Wherever possible, a cushion surface shall be developed on one contact surface. Cement solvents shall be used as recommended by the manufacturer of the particular plastic used and tested by the contractor. Cemented joints shall be free of bubbles and form a strong homogeneous bond. Capillary joints shall require prior written approval of the procuring activity for each specific application.

3.2 Design.- The training aid shall be a three-dimensional mockup of a Pendulous Integrating Gyro Accelerometer (PIGA). The demonstrator shall be sectionalized in part, and the components identified by different colors. Components, such as rotors and gimbals shall rotate freely about their respective axes.

3.2.1 The design is established by the Drawing 3247-040114 and NAVTRADEVCEN Photograph BXA-6983-8-61. The training aid is designed to demonstrate the principles of operation, components arrangement, and the identification of parts in a Pendulous Integrating Gyro Accelerometer (PIGA).

3.3 Reliability.- The training aid shall be designed for a useful classroom life of 20 hours per week for 5 years. Reliability shall be a prime consideration in the design and manufacture of the training aid. The design shall be such as to result in reduced requirements for adjustments or alinement. The training aid shall be constructed so no fixed part shall become free and no moving part shall become loose or sluggish in operation. Degradation shall not be caused because of shipment, normal handling in classroom use or storage. The training aid shall be designed for storage and transportation when under ambient conditions of temperature ranging from minus 5 to plus 125 degrees Fahrenheit, relative humidity ranging up to 95 percent, and still be capable of meeting the specification requirements.

3.4 Construction.- Construction shall be in accordance with processes requirements of MIL-T-23991, and as specified in 3.4.1 and 3.4.2 of this specification.

3.4.1 Breakage and deformation.- The training aid shall be constructed from impact resistant plastic or other suitable material, see 3.1, to minimize the possibility of breakage and deformation during use by the instructor or students.

3.4.2 Bearings, inserts or bushings.- Bearings, inserts or bushings, when they and their support are visible in section, shall be outlined with engraved lines, filled in with black. The bearing shall also be marked by a differentiating color.

3.5 Dimensions and tolerances.-

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3.5.1 Dimensions.- Dimensions not otherwise specified in the drawing or photograph of 3.2, shall be consistent with the functional design of the device and inherent qualities of the material being used.

3.5.2 Tolerances.- Tolerances in construction, not otherwise specified in the drawing or photograph of 3.2 shall be consistent with functional design, inherent qualities of the material being used, and shall afford freedom from binding in structural and mechanical parts, to insure free and easy operation of all moving parts of the demonstrator.

3.6 Details of components.- The training aid shall consist of the following major components:

(a) Mockup

(b) Dust cover.

3.6.1 Mockup.- The mockup of the Pendulous Integrating Gyro Accelerometer shall be a scale model cutaway to show the internal configuration of the Gyro and its components in three dimensional form with color-coded parts.

3.6.1.1 Parts shall be movable manually to demonstrate how the PIGA functions in an inertial guidance stable reference platform to overcome departures from built-in orientation references.

3.6.2 Dust cover.- A dust cover of gray 16 ounce vinyl-coated nylon similar to ARMOR-TITE VN15650 shall be provided to cover the device.

3.6.2.1 The dust cover shall be labeled with black lettering of one-half inch size to identify the device by title and number.

3.7 Finish and color.-

3.7.1 Finish .- Surfaces to be painted shall be properly sealed, primed, and then finished with a satin-surface lacquer in accordance with the Preparation for Painting requirement of MIL-T-23991. Application of finishes shall be accomplished in such a manner as not to impair performance of the training aid.

3.7.1.1 Aluminum parts shall be anodized in accordance with MIL-A-8625.

3.7.2 Colors.- The training aid components, interconnecting parts and fittings, shall be color coded to distinguish fluid, structural and mechanical functions, as indicated on the photograph and drawing of 3.2.

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3.7.2.1 The colors shall be in accordance with color samples furnished by the procuring activity.

3.8 Nameplates and product markings.-

3.8.1 A Government-furnished nameplate and an emblem shall be attached to the outside cylindrical surface of each demonstrator, as indicated on drawings of 3.2.1.

3.9 Workmanship.- The training aid, including all parts and accessories, shall be constructed and finished in a thoroughly workmanlike manner. Particular attention shall be given to neatness and thoroughness of markings and indications of parts and assemblies, painting, and freedom of parts from burrs and sharp edges.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspection.- Inspections to be performed are classified as follows:

(a) In-process inspection

(b) Quality conformance inspection.

4.3 In-process inspection.- In-process inspection shall include such visual and mechanical examination, and testing of materials, subassemblies, parts and accessories (including purchased standard items) during the manufacturing process as may be necessary to assure conformance to all the requirements of this specification.

4.4 Quality conformance inspection.- Quality conformance inspection shall consist of a preproduction inspection and inspection of production units.

4.4.1 Preproduction inspection.- A preproduction inspection shall be conducted on a preproduction unit and shall consist of all the examinations and tests specified in 4.5 and 4.6.

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4.4.2 Production inspection.- Production units shall be inspected to the extent necessary to maintain quality control and conformance to the Government-accepted preproduction unit.

4.5 Examinations.- Examinations shall be performed to verify conformance to the following requirements:

(a) Materials

(b) Design

(c) Construction

(d) Dimensions and tolerances

(e) Details of components

(f) Finish and color

(g) Nameplates and product markings

(h) Workmanship

(i) Preparation for delivery.

4.6 Tests.- Functional tests shall be conducted to determine compliance to the requirements of this specification.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging.- Each training aid shall be preserved and packaged for mechanical and physical protection in accordance with MIL-P-116, Method III.

5.1.1 Unit packaging .- Each training aid shall be packaged one unit to a container conforming to PPP-B-636, Class 2.

5.2 Packing.- Units packaged in accordance with 5.1.1 shall be packed in exterior containers conforming to PPP-B-601, styles A or B with steel banding, or PPP-B-621. The gross weight shall not exceed the weight limitations of the shipping container. Each exterior shipping container shall be lined with a waterproof barrier sealed with adhesive or tape to effect water resistance in accordance with MIL-L-10547. A case liner shall not be required where package(s) and package dunnage have been treated to resist water. Unit fiber boxes, conforming to PPP-B-636, Class 2, exterior shipping containers, closed, reinforced and waterproofed in accordance with the appendix thereto, and used as shipping containers, shall not be overpacked.

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5.3 Physical protection.- All material shall be packaged and packed in containers constructed to provide a compact and nonshifting load. Adequate cushioning, blocking and bracing shall be provided within shipping containers in a manner designed to prevent damage due to shock, vibration or impact during shipment, handling and storage.

5.3.1 Cushioning and covering materials used shall be compatible with all the finishes of the device, so that migrating solvents in cements, cushioning or components will not adversely affect any part.

5.4 Marking.- Interior packages and exterior shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use.- The training aid mockup gyro demonstrator, is intended to be used by instructors as an aid in group training of Naval personnel, or as a self-training medium, in demonstrating the principles and theory of operation, construction, components, and function of the Pendulous Integrating Gyro Accelerometer in an inertial guidance system.

6.2 Ordering data.- Procurement documents should specify the following:

(a) Title, number and date of this specification

(b) Preservation, packaging and packing (see 5.1).

CUSTODIAN:
Navy - TD

PREPARING ACTIVITY:
Navy TD
Project No. 6910-N187

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

SPECIFICATION

ORGANIZATION *(Of submitter)*

CITY AND STATE

CONTRACT NO.

QUANTITY OF ITEMS PROCURED

DOLLAR AMOUNT
$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*

DATE

DD FORM 1 APR 63 1426 REPLACES NAVSHIPS FORM 4863, WHICH IS OBSOLETE

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